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# Solving Quadratics

Stage 1: ERCS — make it = 0. Stage 2: let the calculator decide — clean answers factorise, messy answers go in the formula.

## THE TOPIC ON ONE MAP

### Solving Quadratics

#### 01 — ERCS: make it = 0

$$x(3x - 9) = 4$$

$$x^2 - 6x + 15 = 3x - 5$$

$$\frac{5}{4x + 1} = \frac{2x}{x^2 + 3}$$

ratios

#### 02 — Clean answers: factorise

$$x^2 - 5x - 36 = 0$$

$$5x^2 + 7x + 2 = 0$$

"Solve by factorising"

#### 03 — Messy answers: the formula

"to 2 decimal places"

"to 3 s.f."

$$3x^2 + 4x - 12 = 0$$

find the error

#### 04 — Square root and completing the square

$$3x^2 = 75$$

$$9(x + 3)^2 = 4$$

"in the form  $a \pm \sqrt{b}$ "

"hence solve"

#### 05 — Shapes and hidden quadratics

area  $\Rightarrow$  "show that"

$$6x^4 + 7x^2$$

indices  $\Rightarrow$  quadratic

ratio  $x : y$

**IN THE EXAM** Solving a quadratic is inside about one question in five: graphs, shapes, simultaneous equations.

# 01 ERCS: Make It Equal Zero

You can only solve a quadratic when it is  $ax^2 + bx + c = 0$ . **ERCS** gets you there — then the calculator takes over.

## WHEN YOU SEE → YOU DO

**a bracket:**  $x(3x - 9) = 4$

→ Expand

**terms on both sides**

→ Rearrange: everything to one side,  $= 0$

**fractions or a ratio**  $a : b = c : d$

→ cross-multiply:  $ad = bc$

**“in the form”**  $ax^2 + bx + c = 0$

→ Collect — this line is a mark by itself

Expand

Rearrange ( $= 0$ )

Collect

Solve

### ZONE A

**Answers are whole numbers or fractions**

Factorise (Idea 02).

### ZONE B

**Answers are long decimals**

Quadratic formula (Idea 03).

### ZONE C

**No  $x$  term, or a bracket squared**

Square root both sides —  $\pm!$  (Idea 04)

### ZONE D

**“in the form”**  $(x + a)^2 + b$  / “ $a \pm \sqrt{b}$ ”

Complete the square (Idea 04).

**CALCULATOR CHECK** **MODE 5 3** (equation, degree 2). Type  $a$ ,  $b$ ,  $c$ . The answers tell you which zone you are in.

## WORKED MODEL • RATIO

$2x - 1 : x - 4 = 16x + 1 : 2x - 1$ . Find the possible values of  $x$ .

1  $(2x - 1)^2 = (x - 4)(16x + 1)$

cross-multiply

2  $4x^2 - 4x + 1 = 16x^2 - 63x - 4$

Expand

3  $12x^2 - 59x - 5 = 0$

Rearrange + Collect

4  $(12x + 1)(x - 5) = 0$

Solve (clean answers)

**ANSWER**  $x = 5$  or  $x = -\frac{1}{12}$

## IDEA 01

**Practise: ERCS: Make It Equal Zero**

My questions first, then selected practice.

**Q1** **DR ESLAM** Expand, then = 0

CH 2 · TOPIC 10 3 marks

Solve  $x(x + 3) = 10$

$x =$  \_\_\_\_\_

**SELECTED PRACTICE****Q2** **PRACTICE** Practice · Medium

CH 2 · TOPIC 10 3 marks

13 (a) Write  $x(3x - 9) = 4$  in the form  $ax^2 + bx + c = 0$  where  $a$ ,  $b$  and  $c$  are integers.

(1 mark)

(b) Solve  $x(3x - 9) = 4$

Give your answers to 2 decimal places.

(2 marks)

**Q3****PRACTICE**

Practice · Medium

**CH 2 · TOPIC 10****4 marks****18** Solve.

$$x^2 - 6x + 15 = 3x - 5$$

$$x = \dots\dots\dots \text{ or } x = \dots\dots\dots$$

**(4 marks)****Q4****PRACTICE**

Practice · Hard

**CH 2 · TOPIC 10****4 marks**

**4** Solve  $5x^2 = 10x + 4$

Give your answers to 2 decimal places.

**(4 marks)**

**Q5****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 10

**5 marks****3** Given that

$$2x - 1 : x - 4 = 16x + 1 : 2x - 1$$

find the possible values of  $x$ .**(5 marks)****Q6****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 10

**4 marks****5** Given that

$$x^2 : (3x + 5) = 1 : 2$$

find the possible values of  $x$ .**(4 marks)**

**Q7****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 10

**5 marks**

**10** Solve  $\frac{5}{4x+1} = \frac{2x}{x^2+3}$

Give your solutions to 3 significant figures. You **must** show your working.

**(5 marks)**

## 02 Clean Answers: Factorise

Calculator gives whole numbers or fractions? Then it factorises. Turn each answer into a bracket.

WHEN YOU SEE → YOU DO

“Solve by factorising”

→ brackets are **required**

calculator:  $x = 9, x = -4$

→  $(x - 9)(x + 4) = 0$

calculator:  $x = -\frac{2}{5}$

→  $(5x + 2)$  — the bottom goes in front of  $x$

“the only solution is  $x = 5$ ”

→  $(x - 5)^2 = 0$

MODE 5 3

Answers to brackets

Write = 0

Each bracket = 0

WORKED MODEL · JANUARY 2020 P1HR

Solve  $x^2 - 5x - 36 = 0$ . Show clear algebraic working.

1 calculator:  $x = 9, x = -4$

MODE 5 3

2  $(x - 9)(x + 4) = 0$

answers → brackets

3  $x - 9 = 0$  or  $x + 4 = 0$

each bracket = 0

ANSWER  $x = 9$  or  $x = -4$

× **TRAP** Writing only the calculator answers scores 0 for “show clear algebraic working”. The brackets line is the method mark.

## IDEA 02

**Practise: Clean Answers: Factorise**

My questions first, then selected practice, then real past papers.

**Q8** DR ESLAM Answers to brackets

CH 2 · TOPIC 10 3 marks

Solve  $x^2 - 7x + 12 = 0$ . Show clear algebraic working.

$x =$  \_\_\_\_\_

**SELECTED PRACTICE****Q9** PRACTICE Practice · Medium

CH 2 · TOPIC 10 3 marks

10 (a) Factorise  $x^2 + 2x - 24$ .

(2 marks)

(b) Hence solve  $x^2 + 2x - 24 = 0$

[1]

(1 mark)

**Q10****PRACTICE**

Practice · Medium

CH 2 · TOPIC 10 3 marks

14 Solve  $x^2 - x - 12 = 0$

(3 marks)

**Q11****PRACTICE**

Practice · Hard

CH 2 · TOPIC 10 3 marks

7 Solve by factorisation.

$$5x^2 + 7x + 2 = 0$$

 $x = \dots\dots\dots$  or  $x = \dots\dots\dots$ 

(3 marks)

**Q12****PRACTICE**

Practice · Hard

**CH 2 · TOPIC 10** 3 marks

10 Solve by factorisation  $10r^2 - 23r + 9 = 0$ .

 $r = \dots\dots\dots$  or  $r = \dots\dots\dots$ 

(3 marks)

**PAST PAPERS****Q13****PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 1HR · Q6

**CH 2 · TOPIC 10** 3 marks

6 Solve  $x^2 - 5x - 36 = 0$   
Show clear algebraic working.

.....

(Total for Question 6 is 3 marks)

**Q14** **PAST PAPER** Edexcel 4MA1 · January 2020 · Paper 2H · Q7**CH 2 · TOPIC 10** **5 marks**7 (a) Solve the inequality  $2x + 7 > 4$ (b) Solve  $x^2 - 3x - 40 = 0$   
Show clear algebraic working.**(Total for Question 7 is 5 marks)****Q15** **PAST PAPER** Edexcel 4MA1 · January 2021 · Paper 2HR · Q9**CH 2 · TOPIC 10** **3 marks**9 Solve  $x^2 - 21x + 20 = 0$   
Show your working clearly.**(Total for Question 9 is 3 marks)**

## 03 Messy Answers: The Quadratic Formula

Calculator gives long decimals? It does not factorise. Substitute into the formula and **show** the substitution.

### WHEN YOU SEE → YOU DO

“correct to 2 decimal places / 3 s.f.”

→ quadratic formula

$b$  is negative

→  $-b$  becomes positive:  $-(-4) = 4$

“find the error in her working”

→ check:  $-b, b^2$  in brackets, divide **everything** by  $2a$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Write a b c

Substitute

Calculator

Round

### WORKED MODEL • 2 DECIMAL PLACES

Solve  $3x^2 + 4x - 12 = 0$ , giving your solutions to 2 d.p.

1  $a = 3, b = 4, c = -12$

write  $a, b, c$

2  $x = \frac{-4 \pm \sqrt{4^2 - 4(3)(-12)}}{2(3)} = \frac{-4 \pm \sqrt{160}}{6}$

substitute

ANSWER  $x = 1.44$  or  $x = -2.77$

× **TRAP** The line under the fraction is under **everything**:  $-b$  is divided by  $2a$  too.  $8 \pm \frac{\sqrt{40}}{4}$  is wrong.

IDEA 03

## Practise: Messy Answers: The Formula

My questions first, then selected practice.

**Q16** DR ESLAM Formula

CH 2 · TOPIC 10 3 marks

Solve  $x^2 - 4x - 3 = 0$ , giving your answers to 2 decimal places.

$x =$  \_\_\_\_\_

### SELECTED PRACTICE

**Q17** PRACTICE Practice · Medium

CH 2 · TOPIC 10 3 marks

2 Solve the equation  $3x^2 + 4x - 12 = 0$

Give your solutions correct to 2 decimal places.

(3 marks)

**Q18****PRACTICE**

Practice · Medium

CH 2 · TOPIC 10 3 marks

4 Solve  $3x^2 - x - 1 = 0$

Give your solutions correct to 2 decimal places.

(3 marks)

**Q19****PRACTICE**

Practice · Medium

CH 2 · TOPIC 10 3 marks

7 Solve  $3x^2 - 4x - 2 = 0$

Give your solutions correct to 3 significant figures.

(3 marks)

**Q20****PRACTICE**

Practice · Medium

**CH 2 · TOPIC 10** **3 marks**

8 Solve  $3x^2 - 5x - 1 = 0$

Give your solutions correct to 3 significant figures.

**(3 marks)**

**Q21****PRACTICE**

Practice · Medium

**CH 2 · TOPIC 10** **4 marks**

17 Solve this equation algebraically.

Give your solutions correct to 2 decimal places.

$$3x^2 + 8x - 5 = 0$$

$x = \dots\dots\dots$  or  $x = \dots\dots\dots$

**(4 marks)**

**Q22****PRACTICE**

Practice · Hard

CH 2 · TOPIC 10 6 marks

**6 (a)** Here are two pieces of work.

For each one, describe the error made and give the complete correct solution.

Question:

Solve by factorisation.

$$3x^2 - 2x - 5 = 0$$

Solution:

$$(3x + 5)(x - 1) = 0$$

$$\text{Therefore } x = -5/3 \text{ or } x = 1$$

Error: ..... Correct solution:

**(3 marks)**

Question: Solve, giving your answers correct to 3 significant figures.

$$2x^2 - 8x + 3 = 0$$

Solution:

$$x = -(-8) \pm \frac{\sqrt{(-8)^2 - 4 \times 2 \times 3}}{2 \times 2}$$

$$\text{Therefore } x = 6.42 \text{ or } x = 9.58$$

**(b)**

Error: ..... Correct solution:

**(3 marks)**

## 04 Square Root and Completing the Square

No  $x$  term, or a bracket squared already? Square-root both sides — and never forget the  $\pm$ . Asked for a form? Complete the square.

WHEN YOU SEE → YOU DO

$3x^2 = 75$  (no  $x$  term)

→  $x^2 = 25$ ,  $x = \pm 5$

$9(x+3)^2 = 4$ , “without expanding”

→  $(x+3)^2 = \frac{4}{9}$ ,  $x+3 = \pm \frac{2}{3}$

“in the form  $a \pm \sqrt{b}$ ” / “surd form”

→ complete the square

“use your answer to part (a)” / “hence”

→ use the completed square form

Isolate the square

Square root ( $\pm$ )

Solve both

### WORKED MODEL • SURD FORM

Solve  $x^2 - 6x - 8 = 0$ . Give your answer as  $a \pm \sqrt{b}$ .

1  $(x-3)^2 - 9 - 8 = 0$

complete the square

2  $(x-3)^2 = 17$

isolate the square

3  $x-3 = \pm\sqrt{17}$

square root,  $\pm$

ANSWER  $x = 3 \pm \sqrt{17}$

× **TRAP** Liz wrote  $x^2 = 25 \Rightarrow x = 5$ . She lost  $x = -5$ . Square root always gives **two** answers.

IDEA 04

## Practise: Square Root and Completing the Square

My questions first, then selected practice.

**Q23** DR ESLAM Square root

CH 2 · TOPIC 10 2 marks

Solve  $2x^2 = 50$

$x =$  \_\_\_\_\_

**Q24** DR ESLAM Bracket squared

CH 2 · TOPIC 10 2 marks

Solve  $(x + 3)^2 = 16$  without expanding.

$x =$  \_\_\_\_\_

## SELECTED PRACTICE

**Q25** **PRACTICE** Practice · Medium

CH 2 · TOPIC 10 2 marks

1 Solve  $(x - 2)^2 = 3$

Give your solutions correct to 3 significant figures.

(2 marks)

**Q26** **PRACTICE** Practice · Medium

CH 2 · TOPIC 10 1 mark

6 Liz is asked to solve the equation  $3x^2 + 8 = 83$

Here is her working.

$$3x^2 + 8 = 83$$

$$3x^2 = 75$$

$$x^2 = 25$$

$$x = 5$$

Explain what is wrong with Liz's answer.

(1 mark)

**Q27****PRACTICE**

Practice · Medium

CH 2 · TOPIC 10 2 marks

19 Solve.

$$3x^2 = 75$$

 $x = \dots\dots\dots$ 

(2 marks)

**Q28****PRACTICE**

Practice · Hard

CH 2 · TOPIC 10 3 marks

1 Solve  $x^2 - 6x - 8 = 0$ Write your answer in the form  $a \pm \sqrt{b}$  where  $a$  and  $b$  are integers.

(3 marks)

**Q29****PRACTICE**

Practice · Hard

**CH 2 · TOPIC 10****7 marks**

9 (i) Write  $x^2 + 4x - 16$  in the form  $(x + a)^2 - b$ .

[3]

(ii) Solve the equation  $x^2 + 4x - 16 = 0$ .

Give your answers in surd form as simply as possible.

$x = \dots\dots\dots$  or  $x = \dots\dots\dots$  [4]

**(7 marks)**

**Q30****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 10

**5 marks**

9 (a) Express  $7 - 4x - x^2$  in the form  $p - (x + q)^2$  where  $p$  and  $q$  are constants.

**(2 marks)**

(b) Use your answer to part (a) to solve the equation  $7 - 4(y + 3) - (y + 3)^2 = 0$

Give your solutions in the form  $e \pm \sqrt{f}$  where  $e$  and  $f$  are integers.

**(3 marks)**

## 05 Shapes and Hidden Quadratics

The quadratic is hidden inside an area, a ratio, an index law or a power of  $x^2$ . Build the equation, then ERCS.

### WHEN YOU SEE → YOU DO

“the area is  $351 \text{ cm}^2$ . Show that ...”

→ area formula = 351, then ERCS

“work out the value of  $x$ ” (a length)

→ reject the negative answer

$6x^4 + 7x^2$

→ let  $u = x^2$ :  $6u^2 + 7u$

$6x^2 = 7xy + 20y^2$ , “find  $x : y$ ”

→ factorise:  $(2x - 5y)(3x + 4y) = 0$

### WORKED MODEL · TRAPEZIUM

Parallel sides  $x - 4$  and  $x + 5$ , height  $2x$ . Area = 351. Find  $x$ .

1  $\frac{1}{2}(x - 4 + x + 5)(2x) = 351$

area formula

2  $2x^2 + x - 351 = 0$

Expand + Collect

3  $(2x + 27)(x - 13) = 0$

clean answers

**ANSWER**  $x = 13$  ( $x = -13.5$  is a length  $< 0$ : reject)

**WHERE YOU WILL USE IT** Quadratic inequalities (Topic 11), non-linear simultaneous equations (Topic 12), forming equations (Topic 13).

IDEA 05

## Practise: Shapes and Hidden Quadratics

My questions first, then selected practice.

**Q31** DR ESLAM Rectangle

CH 2 · TOPIC 10 4 marks

A rectangle is  $x$  cm wide and  $(x + 4)$  cm long. Its area is  $60 \text{ cm}^2$ . Find the width.

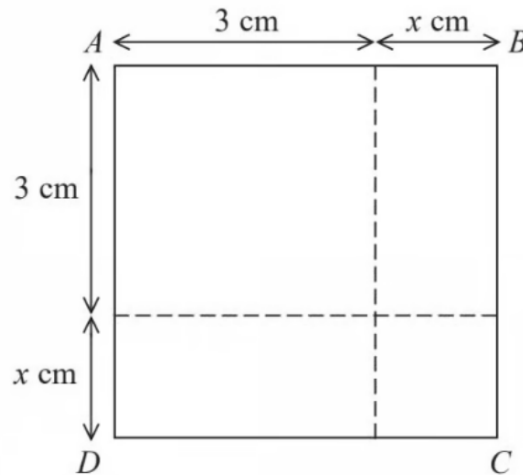
width = \_\_\_\_\_

## SELECTED PRACTICE

**Q32** PRACTICE Practice · Hard

CH 2 · TOPIC 10 3 marks

3



The area of square  $ABCD$  is  $10 \text{ cm}^2$ .

Show that  $x^2 + 6x = 1$

(3 marks)

**Q33****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 10 5 marks

2 (a) Solve  $2x^2 + 9x - 7 = 0$

Give your solutions correct to 3 significant figures.

(3 marks)

(b) Solve  $\frac{2}{y^2} + \frac{9}{y} - 7 = 0$

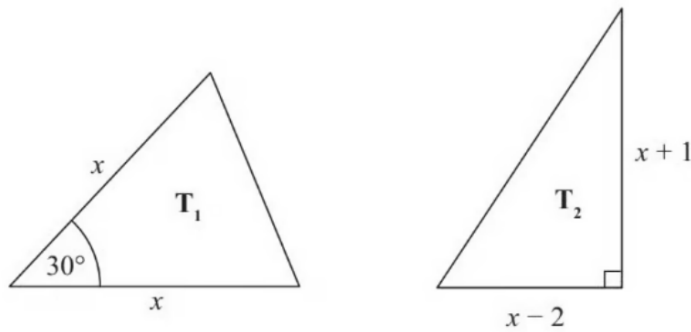
Give your solutions correct to 3 significant figures.

(2 marks)

**Q34** PRACTICE Practice · Very Hard

CH 2 · TOPIC 10 5 marks

6

Diagram **NOT**  
accurately drawn

The lengths of the sides are in centimetres.

The area of triangle **T<sub>1</sub>** is equal to the area of triangle **T<sub>2</sub>**.

Work out the value of  $x$ , giving your answer in the form  $a + \sqrt{b}$  where  $a$  and  $b$  are integers.

**(5 marks)**

**Q35** **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 10 5 marks

7 Given that  $M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of  $n$  for which  $M = 2$

(5 marks)

**Q36** **PRACTICE** Practice · Very Hard

CH 2 · TOPIC 10 6 marks

11  $y = 6x^4 + 7x^2$  and  $x = \sqrt{w+1}$

Find the value of  $w$  when  $y = 10$ . Show your working.

$w = \dots\dots\dots$

(6 marks)

**Q37****PRACTICE**

Practice · Very Hard

CH 2 · TOPIC 10 3 marks

12  $6x^2 = 7xy + 20y^2$  where  $x > 0$  and  $y > 0$

Find the ratio  $x:y$

(3 marks)

# ALL The Whole Topic on One Page

Tear this page out before the exam.



- |           |                      |                                                                                |
|-----------|----------------------|--------------------------------------------------------------------------------|
| <b>01</b> | <b>ERCS</b>          | Expand, Rearrange to = 0, Collect. Ratios and fractions: cross-multiply.       |
| <b>02</b> | <b>Clean answers</b> | Factorise: answers → brackets → each bracket = 0.                              |
| <b>03</b> | <b>Messy answers</b> | Formula; show the substitution; divide <i>everything</i> by $2a$ .             |
| <b>04</b> | <b>Root / CTS</b>    | No $x$ term → square root, $\pm$ . " $a \pm \sqrt{b}$ " → complete the square. |
| <b>05</b> | <b>Hidden</b>        | Area, ratio, $u = x^2$ — build it, ERCS, reject impossible answers.            |

**BOTTOM LINE** ERCS → MODE 5 3 → clean: factorise · messy: formula · a form asked: complete the square.

# Answers

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Solving Quadratics • for the teacher. Every answer checked with a computer algebra system.

## IDEA 01 • ERCS: MAKE IT EQUAL ZERO

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- |           |                             |                                                    |
|-----------|-----------------------------|----------------------------------------------------|
| <b>Q1</b> | Dr Eslam • Expand, then = 0 | $x^2 + 3x - 10 = 0$ ; $x = 2$ or $x = -5$          |
| <b>Q2</b> | Practice • Medium           | (a) $3x^2 - 9x - 4 = 0$ (b) $x = 3.39$ or $-0.39$  |
| <b>Q3</b> | Practice • Medium           | $x^2 - 9x + 20 = 0$ ; $x = 4$ or $5$               |
| <b>Q4</b> | Practice • Hard             | $5x^2 - 10x - 4 = 0$ ; $x = 2.34$ or $-0.34$       |
| <b>Q5</b> | Practice • Very Hard        | $12x^2 - 59x - 5 = 0$ ; $x = 5$ or $-\frac{1}{12}$ |
| <b>Q6</b> | Practice • Very Hard        | $2x^2 - 3x - 5 = 0$ ; $x = 2.5$ or $-1$            |
| <b>Q7</b> | Practice • Very Hard        | $3x^2 + 2x - 15 = 0$ ; $x = 1.93$ or $-2.59$       |

## IDEA 02 • CLEAN ANSWERS: FACTORISE

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- Q8** Dr Eslam · Answers to brackets  $(x - 3)(x - 4) = 0$ ;  $x = 3$  or  $x = 4$
- Q9** Practice · Medium (a)  $(x + 6)(x - 4)$  (b)  $x = -6$  or  $4$
- Q10** Practice · Medium  $(x - 4)(x + 3) = 0$ ;  $x = 4$  or  $-3$
- Q11** Practice · Hard  $(5x + 2)(x + 1) = 0$ ;  $x = -\frac{2}{5}$  or  $-1$
- Q12** Practice · Hard  $(2r - 1)(5r - 9) = 0$ ;  $r = \frac{1}{2}$  or  $\frac{9}{5}$
- Q13** Edexcel 4MA1 · January 2020 · Paper 1HR · Q6  $(x - 9)(x + 4) = 0$ ;  $x = 9$  or  $-4$
- Q14** Edexcel 4MA1 · January 2020 · Paper 2H · Q7 (a)  $x > -1.5$  (b)  $(x - 8)(x + 5) = 0$ ;  $x = 8$  or  $-5$
- Q15** Edexcel 4MA1 · January 2021 · Paper 2HR · Q9  $(x - 1)(x - 20) = 0$ ;  $x = 1$  or  $20$

### IDEA 03 · MESSY ANSWERS: THE FORMULA

- Q16** Dr Eslam · Formula  $x = \frac{4 \pm \sqrt{28}}{2}$ ; 4.65 or -0.65
- Q17** Practice · Medium  $x = 1.44$  or  $-2.77$
- Q18** Practice · Medium  $x = 0.77$  or  $-0.43$
- Q19** Practice · Medium  $x = 1.72$  or  $-0.387$
- Q20** Practice · Medium  $x = 1.85$  or  $-0.180$
- Q21** Practice · Medium  $x = 0.52$  or  $-3.19$
- Q22** Practice · Hard (a) wrong brackets:  $(3x - 5)(x + 1)$ ,  $x = \frac{5}{3}$  or  $-1$   
 (b) 8 not divided by 4:  $x = \frac{8 \pm \sqrt{40}}{4} = 3.58$  or  $0.419$

### IDEA 04 · SQUARE ROOT AND COMPLETING THE SQUARE

- Q23** Dr Eslam • Square root  $x^2 = 25; x = \pm 5$
- Q24** Dr Eslam • Bracket squared  $x + 3 = \pm 4; x = 1 \text{ or } x = -7$
- Q25** Practice • Medium  $x = 2 \pm \sqrt{3}: 3.73 \text{ or } 0.268$
- Q26** Practice • Medium she forgot  $x = -5$  ( $x = \pm 5$ )
- Q27** Practice • Medium  $x = \pm 5$
- Q28** Practice • Hard  $x = 3 \pm \sqrt{17}$
- Q29** Practice • Hard (i)  $(x + 2)^2 - 20$  (ii)  $x = -2 \pm 2\sqrt{5}$
- Q30** Practice • Very Hard (a)  $p = 11, q = 2$  (b)  $y = -5 \pm \sqrt{11}$

### IDEA 05 • SHAPES AND HIDDEN QUADRATICS

- Q31** Dr Eslam • Rectangle  $x^2 + 4x - 60 = 0; (x + 10)(x - 6) = 0; \text{ width } 6 \text{ cm}$
- Q32** Practice • Hard  $(x + 3)^2 = 10 \Rightarrow x^2 + 6x + 9 = 10 \Rightarrow x^2 + 6x = 1$
- Q33** Practice • Very Hard (a)  $x = 0.676 \text{ or } -5.18$  (b)  $y = 1.48 \text{ or } -0.193$
- Q34** Practice • Very Hard  $\frac{1}{4}x^2 = \frac{1}{2}(x + 1)(x - 2); x = 1 + \sqrt{5}$
- Q35** Practice • Very Hard  $3n^2 - 14n - 5 = 0; n = 5 \text{ or } -\frac{1}{3}$
- Q36** Practice • Very Hard  $u = w + 1 = x^2: 6u^2 + 7u - 10 = 0, u = \frac{5}{6}; w = -\frac{1}{6}$
- Q37** Practice • Very Hard  $(2x - 5y)(3x + 4y) = 0; x : y = 5 : 2$